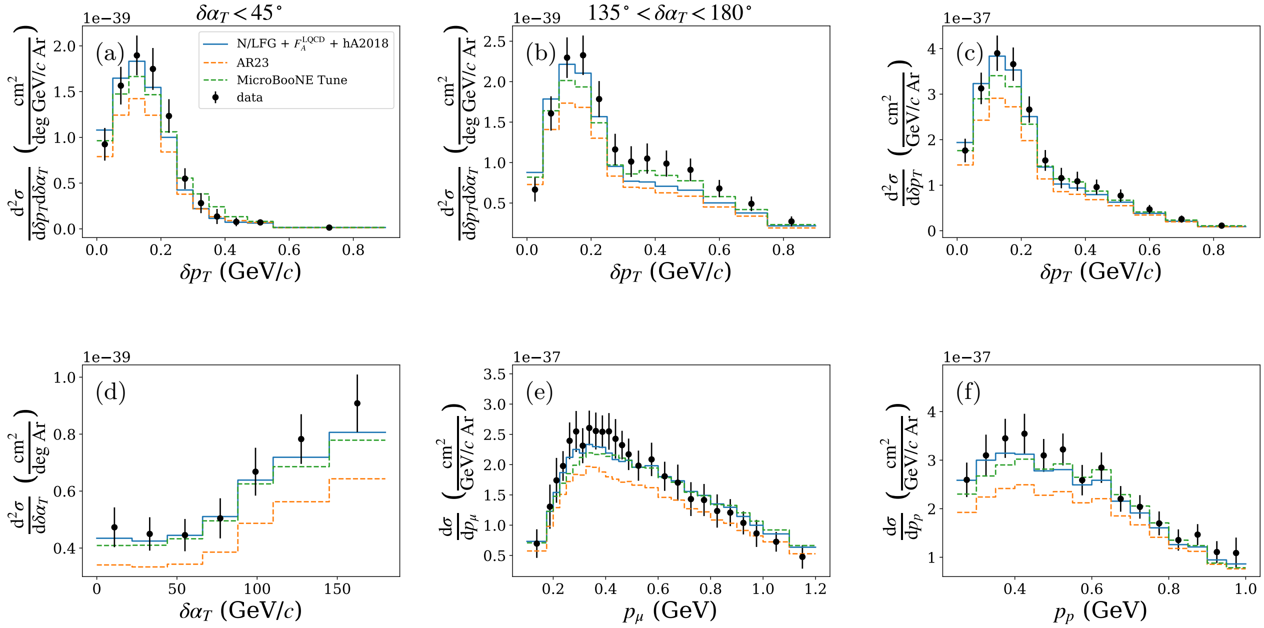

Figure 1: The fully theory-driven GENIE configuration N/LFG + $F_A^{\rm LQCD}$ + hA2018 (blue) compared with the AR23 (orange) and MicroBooNE (green) reference tunes on the MicroBooNE CC1μ1p / CC1μNp argon cross sections (black points). Panels: (a) δp_T for $\delta\alpha_T < 45°$ and (b) for $135° < \delta\alpha_T < 180°$ (two slices of the 2D δp_T–$\delta\alpha_T$ measurement); (c) δp_T; (d) $\delta\alpha_T$; (e) p_μ; (f) p_p. Both reference tunes are drawn dashed. The N/LFG + $F_A^{\rm LQCD}$ + hA2018 prediction gives the lowest χ^2/ndf of the three in every observable (see Table 1).

Table 1: χ^2/ndf (p-value) relative to MicroBooNE data for Fig. 1.

Configuration	δp_T vs. $\delta\alpha_T$	δp_T	$\delta\alpha_T$	p_μ	p_p
N/LFG + $F_A^{\rm LQCD}$ + hA2018	35.98/49 (0.92)	6.78/13 (0.91)	3.47/7 (0.84)	16.29/26 (0.93)	16.05/15 (0.38)
AR23	38.05/49 (0.87)	10.63/13 (0.64)	7.56/7 (0.37)	28.96/26 (0.31)	24.07/15 (0.06)
MicroBooNE Tune	42.61/49 (0.73)	7.31/13 (0.89)	3.96/7 (0.78)	24.59/26 (0.54)	17.38/15 (0.30)